

# MATH1860J Recitation Class 01

Foundations: Chapters 01-05

Shuhan Yu

SJTU Global College

Fall Term 2026

# Outline

1. Introduction
2. Logic
3. Set Theory
4. Natural Numbers and Mathematical Induction
5. Rational Numbers
6. Real Numbers

# 1. Introduction

---

# Introduction

## About me:

- **Shuhan Yu:** Sophomore student majoring in ECE at GC.
- **Email:** barca0927@sjtu.edu.cn
- **Office Hours:**
- **Interest:** Quantitative Trading, Machine Learning, and Geoscience.

## About Honors Mathematics Sequence (8 series):

- Emphasize on rigorous mathematical reasoning and proof.
- Write every step of your solution clearly and logically.
- Don't be frightened by "Honor" in the course title. Remember that 5 series is also an honor series.
- Sample exams are of great help. Go through them before exams.

Feel free to interrupt and ask questions if you have any doubt during RCs!

## 2. Logic

---

## Definition

A **statement** (also called a **proposition**) is a declarative sentence that can be unambiguously classified as either *true* or *false*.

- **Logical operations:**

- Negation (unary operation):  $\neg A$
- Conjunction:  $A \wedge B$
- Disjunction:  $A \vee B$
- Implication:  $A \rightarrow B$
- Converse:  $B \rightarrow A$
- Equivalence:  $A \leftrightarrow B$

Binary operations return a compound statement.

- **Special propositions:**

- Tautology: always true
- Contradiction: always false
- Vacuous truth.

# Propositional Logic

The definition of logical operations is often given by a **truth table**. Consider the implication  $A \rightarrow B$ . It can be rewritten in several equivalent forms (where  $\equiv$  denotes logical equivalence):

$$A \rightarrow B \equiv \neg A \vee B \equiv \neg(A \wedge \neg B) \equiv \neg B \rightarrow \neg A.$$

| $A$ | $B$ | $A \rightarrow B$ | $\neg A \vee B$ | $\neg(A \wedge \neg B)$ | $\neg B \rightarrow \neg A$ |
|-----|-----|-------------------|-----------------|-------------------------|-----------------------------|
| T   | T   | T                 | T               | T                       | T                           |
| T   | F   | F                 | F               | F                       | F                           |
| F   | T   | T                 | T               | T                       | T                           |
| F   | F   | T                 | T               | T                       | T                           |

With the basic operation laws established, we can use them to simplify and analyze more complicated propositions.

# Propositional Logic

## Remark: (Important for Proof)

- $A \rightarrow B \equiv \neg B \rightarrow \neg A$ : equivalence with the **contrapositive**.
- $A \rightarrow B \equiv \neg(A \wedge \neg B)$ : useful in **proof by contradiction**.
- $A \rightarrow B \equiv \neg A \vee B$ : converts the implication into a disjunction, often used in algebraic manipulation.

## Important logical equivalences and laws:

- **Commutativity:**  $p \vee q \equiv q \vee p$ ,  $p \wedge q \equiv q \wedge p$
- **Associativity:**  $(p \vee q) \vee r \equiv p \vee (q \vee r)$
- **Distributivity:**

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

- **De Morgan's Laws:**

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$



# Predicate Logic

A **statement** can be expressed as:

**statement** = **quantifier** + **predicate (statement frame)**.

## Definition

A **predicate** is an expression  $E(x)$  that becomes a statement when  $x$  is replaced by an object from a **specified class** of objects.

- **Universal quantifier**  $\forall$ : “for all”. The expression  $\forall x \in X : E(x)$  means “Every object  $x$  in  $X$  satisfies property  $E$ ”.
- **Existential quantifier**  $\exists$ : “there exists”. The expression  $\exists x \in X : E(x)$  means “There exists at least one object  $x$  in  $X$  with property  $E$ ”.
- **Unique existential**  $\exists!$ : exactly one such object exists.

In all cases, **the order of the quantifiers is significant**. Thus,

$$\forall x \exists y : E(x, y) \not\Leftrightarrow \exists y \forall x : E(x, y)$$

- Former, for all  $x$  there exists some  $y$  such that  $E(x, y)$  is true.
- Latter, there exists a fixed  $y$  such that  $E(x, y)$  is true for all  $x$ .

# Exercise

## Exercise 01

- 1 Show that

$$(A \wedge B) \rightarrow C \equiv A \rightarrow B \rightarrow C \equiv B \rightarrow A \rightarrow C.$$

- 2 Give the negations of the statements:

$$\forall x \in X, \exists y \in Y : E(x, y, z)$$

$$\forall \epsilon > 0, \exists \delta > 0 : |x - a| < \delta \implies |f(x) - f(a)| < \epsilon$$

### Remarks:

Using the quantifiers  $\exists$  and  $\forall$ , negation becomes a purely “mechanical” process in which the symbols  $\exists$  and  $\forall$  (as well as  $\wedge$  and  $\vee$ ) are interchanged (without changing the order), and statements are negated.

### 3. Set Theory

---

# Set Basics and Operations

Sets are collections of objects, called **elements** or **members**.

- Define by Listing:  $A = \{1, 2, 3, 4\}$
- Define by Predicate:  $A = \{x \in \mathbb{N} \mid 1 \leq x \leq 4\}$
- Set  $A = B$  iff  $A \subseteq B$  and  $B \subseteq A$ . (Useful in Proof)

Let  $A = \{x \mid P_1(x)\}$  and  $B = \{x \mid P_2(x)\}$ .

- Union:  $A \cup B := \{x \mid P_1(x) \vee P_2(x)\}$
- Intersection:  $A \cap B := \{x \mid P_1(x) \wedge P_2(x)\}$
- Difference:  $A \setminus B := \{x \mid P_1(x) \wedge \neg P_2(x)\}$ .
- Complement:  $A^c(\overline{A}) := \mathbb{U} \setminus A := \{x \in \mathbb{U} \mid \neg P_1(x)\}$ .

If  $A \cap B = \emptyset$ , we say that  $A$  and  $B$  are **disjoint**.

**Remark:** Distinguish between **Difference**( $A \setminus B$ ) and **Quotient**( $A/B$ )

## Exercise 02

Verify the following set-theoretic identities:

- ①  $(A \cup B)^c = A^c \cap B^c$
- ②  $(A \cap B)^c = A^c \cup B^c$
- ③  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- ④  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- ⑤  $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$
- ⑥  $(A \cap B) \setminus C = (A \setminus C) \cap (B \setminus C)$
- ⑦  $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
- ⑧  $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

**Hint:** Use element-wise reasoning and draw a Venn diagram.

# Power Set and Cardinality

- **Power Set:**  $P(A) = \{X \mid X \subseteq A\}$
- **Cardinality:**  $|A|$  is the number of elements in set  $A$
- **Finite Set:** A set with a finite number of elements
- **Infinite Set:** A set with an infinite number of elements
- **Countable Set:** A set that can be put into one-to-one correspondence with the natural numbers
- **Uncountable Set:** A set that cannot be put into one-to-one correspondence with the natural numbers
- **Cardinality of Power Set:**  $|P(A)| = 2^{|A|}$  if  $A$  is finite

# Ordered Pairs and Cartesian Product

- **Ordered Pair:**  $(a, b)$
- **Cartesian Product:**  $A \times B = \{(a, b) \mid a \in A, b \in B\}$
- **n-tuple:** generalization of ordered pair,  $(a_1, a_2, \dots, a_n)$
- Construction of ordered pair and natural numbers:  
 $(a, b) = \{\{a\}, \{a, b\}\}; 0 = \{\emptyset\}, 1 = \{\emptyset\}, 2 = \{\emptyset, \{\emptyset\}\}, \dots$

- Note that  $(a, b) = (c, d) \iff a = c \wedge b = d$ .
- In general,  $A \times B \neq B \times A$ .



# Naive Set Theory and Russell's Paradox

**Naive Set Theory:** You can define a set by any property. A set is any collection of distinct objects. **Problem: Russell's Paradox** arises when considering the set of all sets that do not contain themselves:

$$S = \{x \mid x \text{ is a set and } x \notin x\}$$

**Question:** Is  $S \in S$ ?

- If  $S \in S$ , then by definition  $S \notin S$ .
- If  $S \notin S$ , then by definition  $S \in S$ .

**Conclusion:** Contradiction arises. Not every property can define a set freely.  $\Rightarrow$  **Implication:** Motivates axiomatic set theory (e.g., Zermelo-Fraenkel Set Theory, ZFC) to avoid such paradoxes.

# Cantor's Paradox (Supplement)

**Prove that “the set of all sets” does not exist.**

**Cantor's Theorem:** For any set  $A$ ,

$$|A| < |P(A)|,$$

where  $P(A)$  is the power set of  $A$ , i.e., the set of all subsets of  $A$ .

**Consider:** The “set of all sets”  $U$ .

- By definition,  $U$  contains every set, so  $P(U) \subseteq U$ .
- By Cantor's Theorem,  $|U| < |P(U)|$ .

**Contradiction:**  $|U| \geq |P(U)|$  from definition, but  $|U| < |P(U)|$  by Cantor.

**Conclusion:** There is no “set of all sets”. This prevents the paradox.

# Zermelo-Fraenkel Set Theory (ZF)

- ➊ **Axiom of Extensionality:** Two sets are equal if they have the same elements.
- ➋ **Axiom of Empty Set:** There exists a set with no elements, denoted  $\emptyset$ .
- ➌ **Axiom of Pairing:** For any sets  $a$  and  $b$ , there exists a set  $\{a, b\}$ .
- ➍ **Axiom of Union:** For any set of sets  $A$ , there exists a set  $\bigcup A$  containing all elements of elements of  $A$ .
- ➎ **Axiom of Power Set:** For any set  $A$ , there exists a set  $P(A)$  of all subsets of  $A$ .
- ➏ **Axiom of Infinity:** There exists a set containing  $\emptyset$  and closed under successor.
- ➐ **Axiom of Replacement:** The image of a set under any definable function is also a set.
- ➑ **Axiom of Separation (Specification):** For any set  $A$  and property  $P$ ,  $\{x \in A \mid P(x)\}$  is a set.
- ➒ **Axiom of Regularity (Foundation):** Every non-empty set  $A$  contains an element disjoint from  $A$ .

# Exercise

## Exercise 03

Let  $A, B, C$  be three statements. Use truth table to prove that

$$(A \vee B) \wedge C \equiv (A \wedge C) \vee (B \wedge C)$$

Let  $A, B, C$  be three sets. Prove that (may use Venn Diagram)

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

## 4. Natural Numbers and Mathematical Induction

---

# Natural Numbers

## Natural Numbers from ZF Set Theory

- By the **Axiom of Infinity**, there exists a set  $I$  such that:
  - $\emptyset \in I$
  - For each  $x \in I$ ,  $S(x) := x \cup \{x\} \in I$
- **Define** natural numbers **recursively**:

$$0 := \emptyset, \quad 1 := S(0) = \{\emptyset\}, \quad 2 := S(1) = \{\emptyset, \{\emptyset\}\}, \dots$$

- The set of natural numbers  $\mathbb{N}$  is the smallest set containing 0 and closed under  $S$ .
- Ordering: The natural numbers are well-ordered.

**Comment:**  $\{\cdot\}$  is packaging and  $\cup$  is stacking.

Each new number encompasses all its previous history and incorporates “the previous self” as a new element. This is why natural numbers have sequentiality.

## Recursive Definition of Addition on $\mathbb{N}$

- **Base case:**

$$\forall n \in \mathbb{N}, \quad n + 0 := n$$

- **Successor step:**

$$\forall n, m \in \mathbb{N}, \quad n + S(m) := S(n + m)$$

*Explanation:*  $S(m) = m \cup \{m\}$  is the successor of  $m$ , so addition recursively “adds 1” using the successor operation.

## Properties of addition:

- Commutativity:  $n + m = m + n$
- Associativity:  $(n + m) + k = n + (m + k)$
- Identity/Existence of a neutral element:  $n + 0 = n$

# Natural Numbers

## Recursive Definition of Multiplication on $\mathbb{N}$

- **Base case:**

$$\forall n \in \mathbb{N}, \quad n \cdot 0 := 0$$

- **Successor step:**

$$\forall n, m \in \mathbb{N}, \quad n \cdot S(m) := (n \cdot m) + n$$

*Explanation:* Multiplication is defined as repeated addition, consistent with the intuitive meaning of natural numbers.

## Properties of multiplication:

- Commutativity:  $n \cdot m = m \cdot n$
- Associativity:  $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- Identity/Existence of a neutral element:  $n \cdot 1 = n$
- Distributivity:  $n \cdot (m + k) = (n \cdot m) + (n \cdot k)$



## Definition (Principle of Mathematical Induction)

Let  $A(n)$  be a statement depending on  $n \in \mathbb{N}$  and let  $n_0 \in \mathbb{N}$ . To prove  $A(n)$  for all  $n \geq n_0$ , it suffices to verify:

- 1 **Base Case:**  $A(n_0)$  is true.
- 2 **Inductive Step:** For all  $n \geq n_0$ , if  $A(n)$  is true, then  $A(n+1)$  is true:

$$\forall n \in \mathbb{N}, n \geq n_0 : A(n) \implies A(n+1).$$

**Conclusion:** If both conditions hold,  $A(n)$  is true for all  $n \geq n_0$ .

**Remark:** The principle of induction is essentially part of the definition of the natural numbers  $\mathbb{N}$ .

# Mathematical Induction

## Exercise 04

Suppose  $M$  is a finite set of men.

**Proposition:** For any  $a, b \in M$ ,  $a = b$ .

**Proof (by induction on  $|M|$ ):**

- ① **Base Case:** If  $|M| = 1$ , the claim is trivially true.
- ② **Inductive Step:** Assume true for  $n$  men. Let  $|M| = n + 1$  and  $a, b \in M$ . Define  $M_a = M \setminus \{a\}$ ,  $M_b = M \setminus \{b\}$ . Take  $c \in M_a \cap M_b$ . Then  $a = c$ ,  $b = c \implies a = b$ .

**Question:** Where is the flaw in this proof?

## Exercise 05

Use mathematical induction, find the general term formula for the Fibonacci sequence defined recursively by:

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2.$$

## 5. Rational Numbers

---

# Rational Numbers and Basic Properties

The set of rational numbers is introduced from set theory as

$$\mathbb{Q} := \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Recap the basic properties of rational numbers:

## Properties for Addition:

- Commutativity:  $a + b = b + a$
- Associativity:  
 $(a + b) + c = a + (b + c)$
- Existence of additive identity:  
 $a + 0 = a$
- Existence of additive inverse:  
 $a + (-a) = 0$

## Properties for Multiplication:

- Commutativity:  $a \cdot b = b \cdot a$
- Associativity:  
 $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- Existence of multiplicative identity:  $a \cdot 1 = a$
- Existence of multiplicative inverse:  $a \cdot a^{-1} = 1$  for  $a \neq 0$
- Distributivity:  
 $a \cdot (b + c) = a \cdot b + a \cdot c$

# Absolute Value and Triangle Inequality

## Definition (Absolute Value)

The **absolute value** (also called the **modulus**) of a rational number  $x$  is defined as

$$|x| := \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

## Theorem (Triangle Inequality)

For any  $x, y \in \mathbb{Q}$ , the following inequality holds:

$$|x + y| \leq |x| + |y|.$$

*Inverse Triangle Inequality:*

$$||x| - |y|| \leq |x - y|.$$

# Trichotomy Law and Closedness

We assume that we know what a **strictly positive** rational number is, i.e., that there exists a set  $P$  with the property that for every number  $a \in \mathbb{Q}$ , one and only one of the following holds:

- 1  $a \in P$  (i.e.,  $a$  is strictly positive),
- 2  $-a \in P$  (i.e.,  $a$  is strictly negative),
- 3  $a = 0$ .

This is known as the **trichotomy law** for rational numbers (**P10**). It allows us to compare any two rational numbers and determine their order.

We also assume that the set of positive numbers is **closed** under addition and multiplication:

- 1 If  $a, b \in P$ , then  $a + b \in P$ . (**P11**)
- 2 If  $a, b \in P$ , then  $a \cdot b \in P$ . (**P12**)

## 6. Real Numbers

---

# Bounds and Boundedness

## Definition (Upper and Lower Bounds)

Let  $(X, \leq)$  be a partially ordered set and  $A \subseteq X$  be nonempty.

- An element  $s \in X$  is an **upper bound** of  $A$  if  $a \leq s$  for all  $a \in A$ .
- An element  $s \in X$  is a **lower bound** of  $A$  if  $a \geq s$  for all  $a \in A$ .

## Definition (Bounded Sets in $\mathbb{Q}$ )

A set  $U \subset \mathbb{Q}$  is said to be **bounded** if there exists a constant  $c \in \mathbb{Q}$  such that

$$|x| \leq c \text{ for all } x \in U.$$

If this is not the case, we say that  $U$  is **unbounded**.

We say that a set  $U \subset \mathbb{Q}$  is **bounded above** if there exists an upper bound for  $U$ , and **bounded below** if there exists a lower bound for  $U$ .



# Maximum and Minimum, Supremum and Infimum

## Definition (Maximum and Minimum)

Let  $U \subset \mathbb{Q}$ .

- $x_1 \in U$  is the **minimum** of  $U$  ( $x_1 = \min U$ ), if  $x_1 \leq x$  for all  $x \in U$ .
- $x_2 \in U$  is the **maximum** of  $U$  ( $x_2 = \max U$ ), if  $x_2 \geq x$  for all  $x \in U$ .

**Note:** A set not bounded below has no minimum, and a set not bounded above has no maximum. Even a bounded set may not have a max or min.

## Definition (Supremum and Infimum)

Let  $U \subset \mathbb{Q}$ .

- An upper bound  $c_2 \in \mathbb{Q}$  of  $U$  is the **least upper bound** or **supremum**, denoted  $c_2 = \sup U$ , if no  $c \in \mathbb{Q}$  with  $c < c_2$  is an upper bound.
- A lower bound  $c_1 \in \mathbb{Q}$  of  $U$  is the **greatest lower bound** or **infimum**, denoted  $c_1 = \inf U$ , if no  $c \in \mathbb{Q}$  with  $c > c_1$  is a lower bound.

# Greatest Lower and Least Upper Bounds of Sets

## Definition

We say that an upper bound  $c_2 \in \mathbb{Q}$  of a set  $U \subset \mathbb{Q}$  is the **least upper bound** or **supremum** of  $U$  if no  $c \in \mathbb{Q}$  with  $c < c_2$  is an upper bound. We then write  $c_2 =: \sup U$ .

We say that a lower bound  $c_1 \in \mathbb{Q}$  is the **greatest lower bound** or **infimum** of  $U$  if no  $c \in \mathbb{Q}$  with  $c > c_1$  is a lower bound. We then write  $c_1 =: \inf U$ .

**Problem:** Rational numbers are insufficient.



We define the set of real numbers  $\mathbb{R}$  as the smallest extension of the rational numbers  $\mathbb{Q}$  such that the following property holds:

## Least Upper Bound Property (P13)

If  $A \subset \mathbb{R}$ ,  $A \neq \emptyset$ , and  $A$  is bounded above, then there exists a least upper bound for  $A$  in  $\mathbb{R}$ .

We call all real numbers that are not rational “**irrational numbers**”.

# Subsets of the Real Numbers

## Definition

Let  $A \subset \mathbb{R}$  be any set.

- ❶ We call  $x \in \mathbb{R}$  an **interior point** of  $A$  if there exists some  $\varepsilon > 0$  such that the interval  $(x - \varepsilon, x + \varepsilon) \subset A$ . The set of all interior points of  $A$  is denoted by  $\text{int } A$ .
- ❷ We call  $x \in \mathbb{R}$  an **exterior point** of  $A$  if there exists some  $\varepsilon > 0$  such that  $(x - \varepsilon, x + \varepsilon) \cap A = \emptyset$ .
- ❸ We call  $x \in \mathbb{R}$  a **boundary point** of  $A$  if for every  $\varepsilon > 0$ , both  $(x - \varepsilon, x + \varepsilon) \cap A \neq \emptyset$  and  $(x - \varepsilon, x + \varepsilon) \cap A^c \neq \emptyset$ . The set of all boundary points of  $A$  is denoted by  $\partial A$ .
- ❹ We call  $x \in \mathbb{R}$  an **accumulation point** of  $A$  if for every  $\varepsilon > 0$ ,  $(x - \varepsilon, x + \varepsilon) \cap (A \setminus \{x\}) \neq \emptyset$ .

# Subsets of the Real Numbers

## Definition

- ❶ A set  $A \subset \mathbb{R}$  is called **open** if all points of  $A$  are interior points; that is, if  $A = \text{int } A$ .
- ❷ A set  $A \subset \mathbb{R}$  is called **closed** if its complement  $\mathbb{R} \setminus A$  is open.
- ❸ The set

$$\overline{A} := A \cup \partial A$$

is called the **closure** of  $A$ . It can be shown that  $\overline{A}$  is the smallest closed set that contains  $A$ .

## Exercise 06

Show that  $\mathbb{R} = \text{int } A \cup \text{ext } A \cup \partial A$ , and think about the sets that are both open and closed.

## Exercise 07

Proof that:

- a For any collection  $\{G_\alpha\}$  of open sets, the union  $\bigcup_\alpha G_\alpha$  is open.
- b For any collection  $\{F_\alpha\}$  of closed sets, the intersection  $\bigcap_\alpha F_\alpha$  is closed. (2.21)
- c For any finite collection  $G_1, G_2, \dots, G_n$  of open sets, the intersection  $\bigcap_{i=1}^n G_i$  is open.
- d For any finite collection  $F_1, F_2, \dots, F_n$  of closed sets, the union  $\bigcup_{i=1}^n F_i$  is closed.

## References:

- Horst Hohberger, *math186\_all\_lecture\_slides (2026)*
- Wenxin Chen, *MATH1860J RC*, 25 Fall
- Leqi Tang, *MATH1860J RC*, 25 Fall

## To be continued...

- Complex Numbers
- Functions and Maps
- Sequences

# Thanks for Listening!

Any questions? Please feel free to ask!